

Customer Analytics Report

CUSTOMER SHOPPING BEHAVIOR ANALYSIS

Data Pipeline Project & Behavioral Insights



Dataset: 3,900 Records × 18 Attributes



Segmentation, Analytics & Database Integration



PROJECT OVERVIEW & EXECUTIVE SUMMARY

Customer Shopping Behavior Analysis

Analyzing 3,900 customer records to uncover shopping habits and key purchase drivers including discounts, reviews, seasonal trends, and payment preferences for advanced analytics and customer segmentation.





DATASET AT A GLANCE

3,900 Rows × 18 Initial Columns
Key Customer & Shopping Attributes

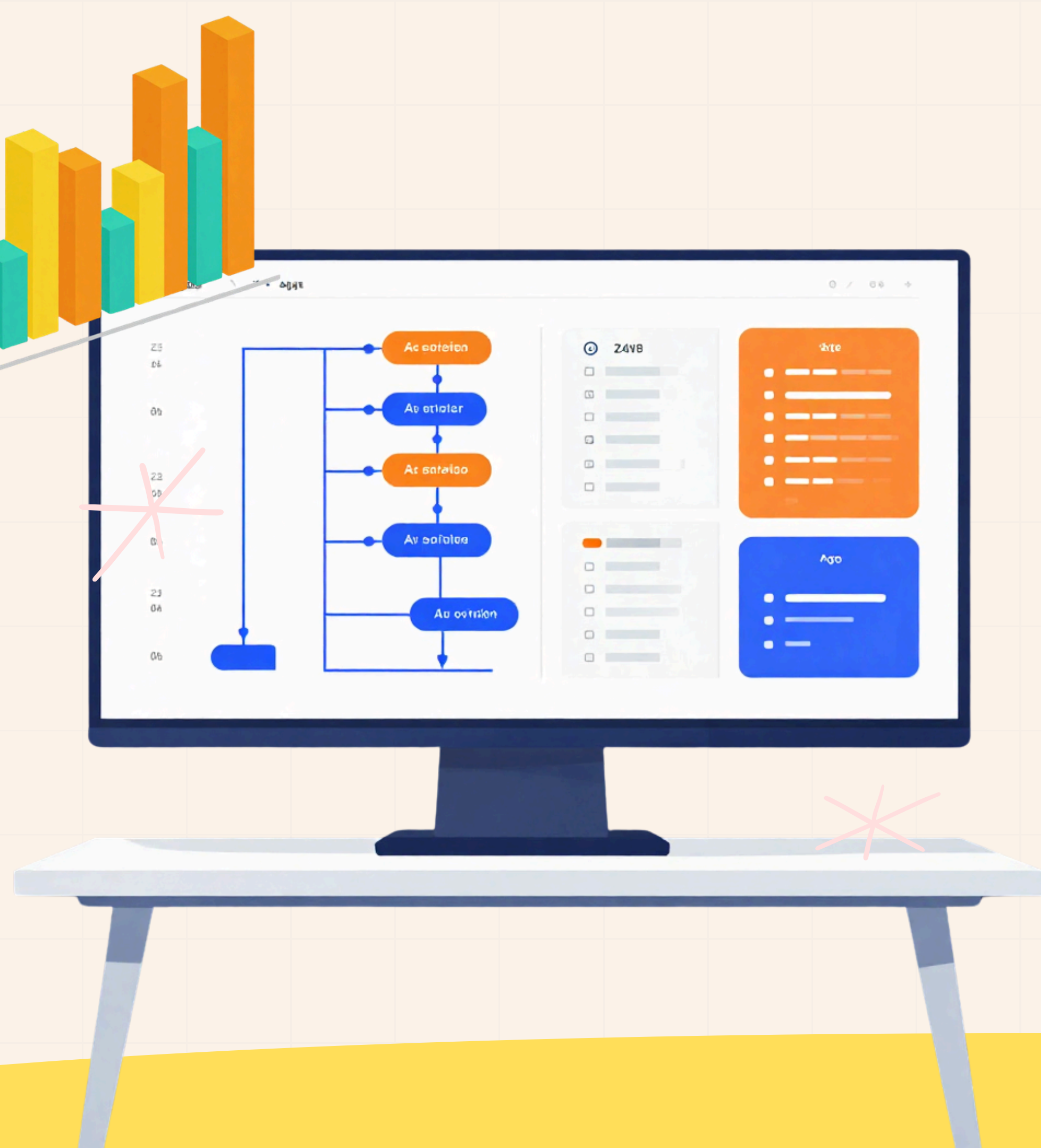
- 3,900 customer transaction records collected
- 18 raw columns covering demographics & behavior
- Attributes: age, gender, category, purchase amount
- Includes reviews, discounts, shipping & payment info
- Seasonal and subscription status data included

DATA CLEANING PIPELINE

Transforming Raw Data into Reliable Insights

- Detection → Identify missing review ratings and nulls
- Imputation → Category-based fill for missing values
- Standardization → Uniform column naming conventions
- Normalization → Scale numeric fields to consistent ranges
- Validation → Verify data integrity and completeness





FEATURE ENGINEERING

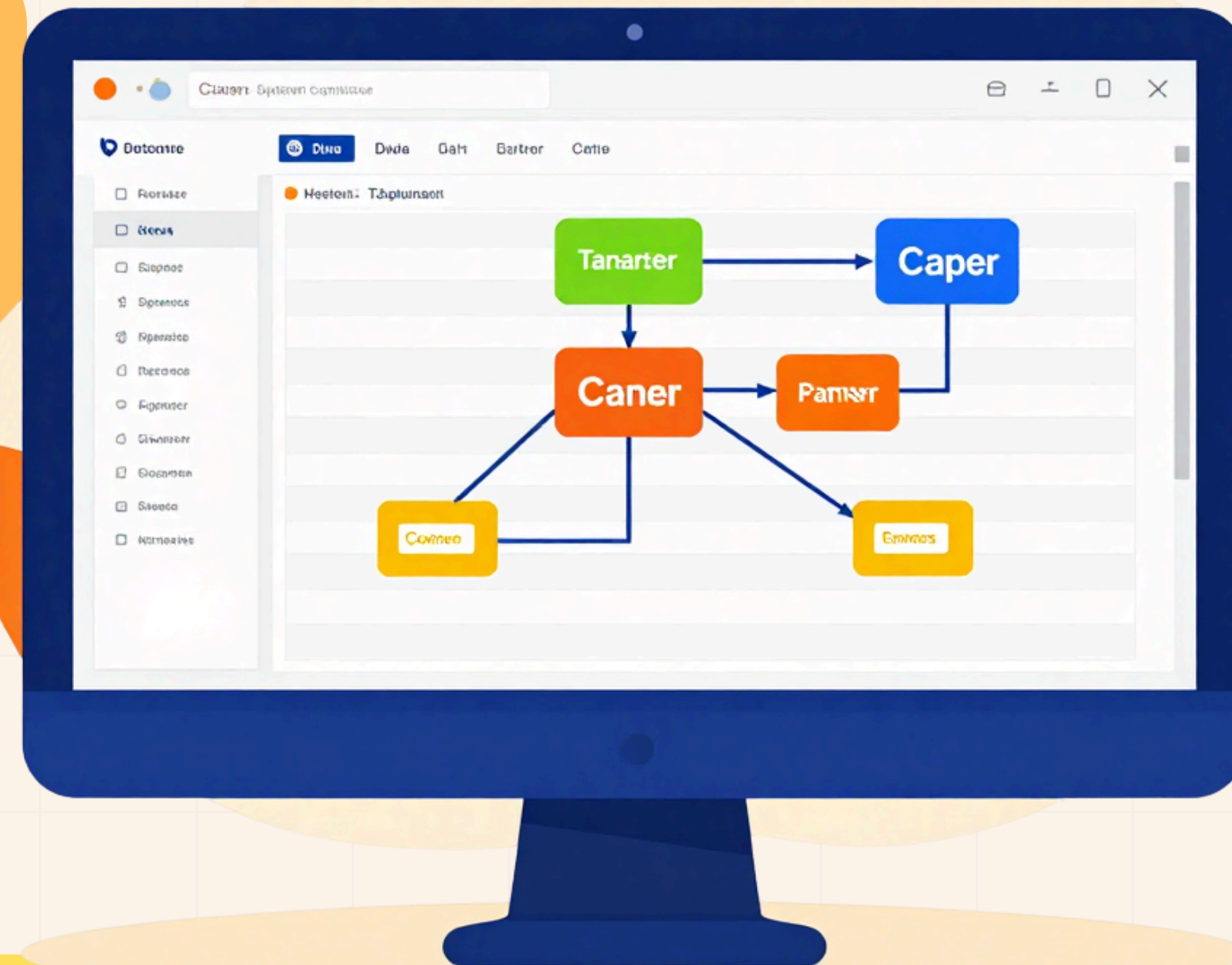
Transforming Raw Data into Meaningful Features

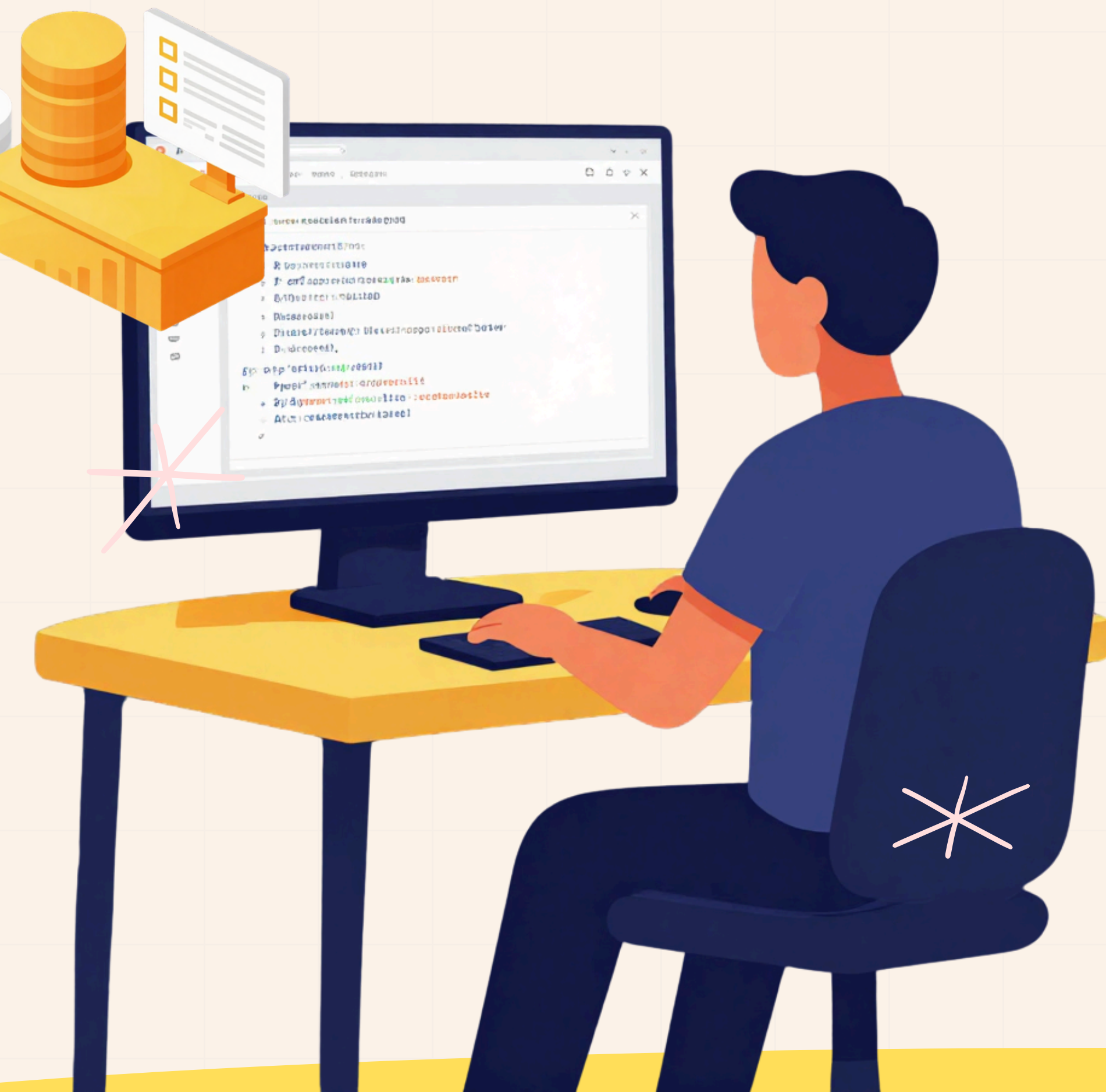
- Age-group binning for demographic segmentation
- Purchase-frequency conversion to numeric scale
- Categorical mapping for machine-ready encoding
- Redundancy reduction to streamline features
- Engineered schema optimized for analytics

DATABASE INTEGRATION

PostgreSQL via SQLAlchemy & psycopg2

Customer data loaded into a PostgreSQL database using SQLAlchemy and psycopg2. A structured customer table stores the final 19-column schema, enabling efficient querying and business analysis.

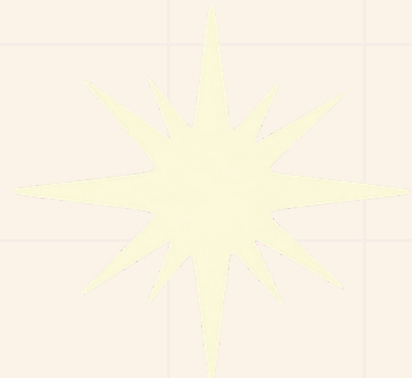




SQL BUSINESS ANALYSIS

Querying Key Business Metrics from Customer Data

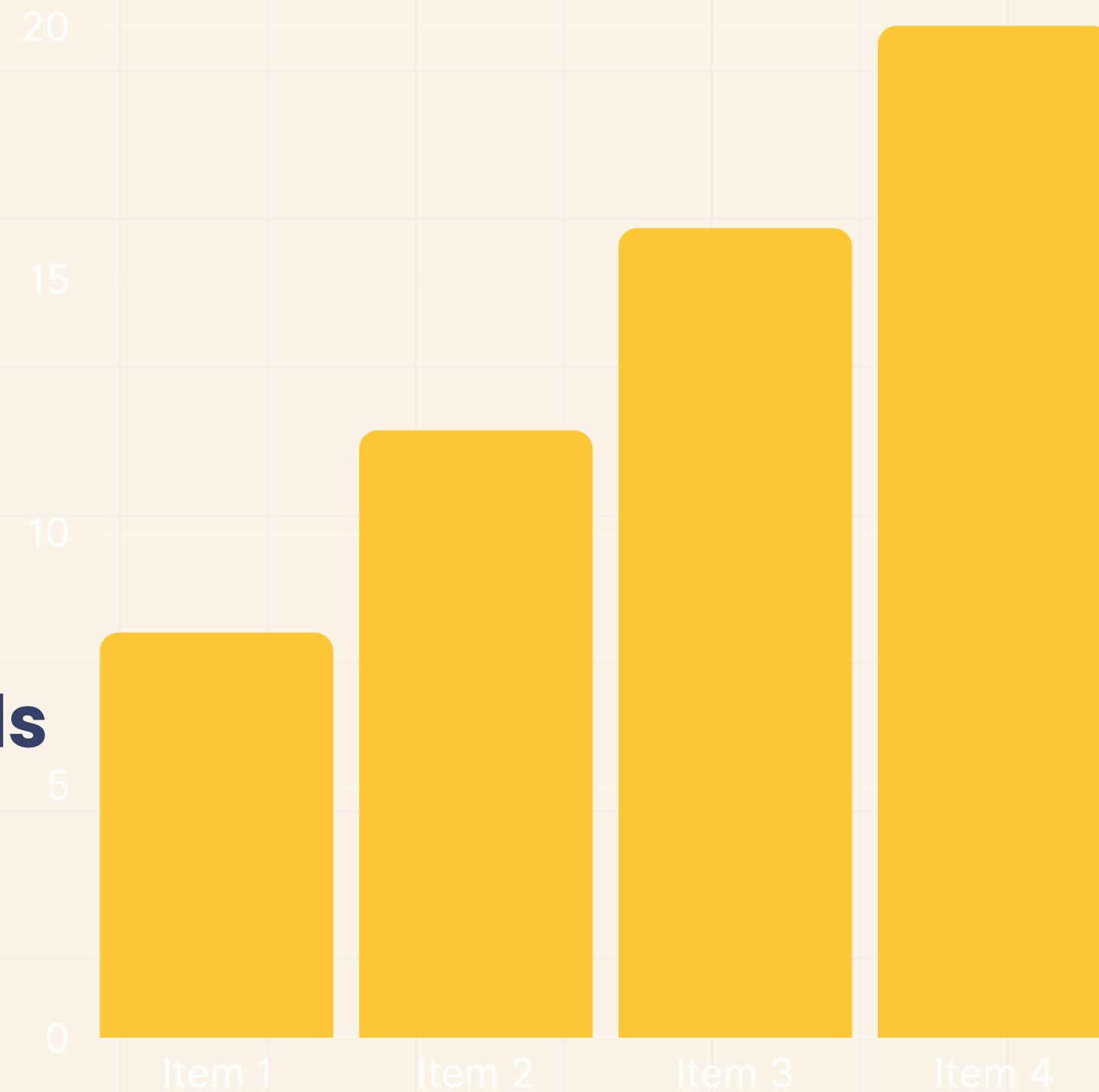
- Revenue breakdown by gender segment
- High-spending discount users identified
- Highly rated products by category
- Shipping method performance comparison
- Subscriber vs. non-subscriber & discount-dependent products



CUSTOMER & PRODUCT INSIGHTS

Segments, Products & Revenue Trends

New, returning, and loyal customers show distinct value patterns. Top 3 products per category drive category revenue. Subscription adoption is growing, and revenue by age group reveals key demographic opportunities.





CONCLUSIONS

Key Findings from the Shopping Behavior Analysis

- Behavior varies by demographics, spending & frequency
- Discounts are a critical driver of purchase decisions
- Returning & loyal customers deliver the highest value
- Product demand and repurchase timing show clear patterns
- Subscription potential remains a strong growth opportunity



BUSINESS RECOMMENDATIONS

Actionable Strategies for Growth

Target age/behavior segments with tailored campaigns, deliver personalized promotions, send automated repurchase reminders, optimize discount strategies, highlight top-rated products, launch loyalty & subscription offers, and maintain continuous data analysis.





THANK YOU & LET'S CONNECT

**Data-Driven Customer Engagement
& Retention Starts Here**



Customer Shopping Behavior Analysis



Data Pipeline Project Full Report



3,900 Records | 19-Column Schema



PostgreSQL · Python · SQLAlchemy

